# Supplementary Material S4
# Complete Stratified Synthesis of the Corpus

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document complements Table 17 in the manuscript (studies of greatest translational traceability, Section 6.3). Section A presents the 36 eligible studies organised by clinical domain (D1–D4); Section B presents the 9 contextual studies outside the eligible set; Section C presents the systematic longitudinal tracking audit and the multi-label coding matrix for all 36 eligible studies.

The internal # column corresponds to the corpus order shared with Supplementary Material S2 (C prefix for contextual evidence), not to bibliographic reference numbers. Gaps in the sequence are not numbering errors: numbers were assigned before the final integrity and eligibility verification and are not reused. Ahmad et al. (2024), retracted after corpus finalisation (17 August 2026, *Healthc Technol Lett* 2026;13(1):e70095), is excluded with retroactive effect. The 8 retracted or withdrawn records do not appear in this document.

Version V3.6 reconstructed the screening (Supplementary Material S5). Seventeen studies were added (identifiers 29 to 51); their coding is first-reviewer coding pending confirmation by a second reviewer, and ND marks fields that could not be confirmed in the full text. Reed et al. (2021) and Almeida et al. (2019) left the eligible corpus because they do not use or evaluate an automated analysis of the face or gaze (criterion CI(1) made explicit in V3.6).

## Section A. Eligible corpus (36 studies)

| # | Study | Clinical task | Dataset | Population / Setting | Architecture | Metrics | Data prov. |
|---|---|---|---|---|---|---|---|
| *Domain D1: Automated facial expression (12 studies)* | | | | | | | |
| 1 | Johnson et al. (2025) | Emotion recogn. | AffectNet + clinical | ASD children ($N$=48) / Lab./Ecol. | EfficientNet-v2 | ACC 67.8%[k] | Clinical |
| 8 | Radočaj & Mart. (2025) | Emotion recogn. | FERAC | ASD children / Lab. | Swin Transformer | ACC 80%, F1 78.9% | Partial prov. |
| 9 | Washington et al. (2022) | Emotion recogn. | CAFE Sub. A + clinical | Children / Lab./Ecol. | ResNet-152 | Bal.ACC 79.1%, F1 78% | Clinical |
| 10 | Sholikah et al. (2024) | Real-time recogn. | FER-2013 | $N$=3 ASD (test)[a] / Mobile | VGG-16 | ACC 91%, F1 91% | Public gen. |
| 21 | Aljbori et al. (2025) | Emotion recogn. | Kaggle ASD | ASD children / Lab. | 3D-CNN + LSTM | ACC 99.01%[i], F1 98.6% | Web |
| 27 | Dapogny et al. (2019) | Facial expr. | JEMImE ($N$=157) | Children 6–11 / Clin. game | Rand. Forests + PCRF | ACC 81.9% | Public cur. |
| 28 | Leo et al. (2018) | Facial prod. ASD | CK+ + clinical ($N$=17) | ASD children / Clinical | HOG + SVM + CNN | TP 85% (CK+)[b] | Clinical |
| 29 | Kiejdo et al. (2025) | Emotion recogn. | EMBOA (99 recordings) | ASD 0–12 / Robot th. | BiLSTM + late fusion | ACC 68%, F1 68% | Clinical |
| 30 | Wang et al. (2025) | Emotion recogn. | Private ($N$=33) | ASD 4–5 / Rehab. classroom | CvT + MS-CAM fusion | ACC 90.73% | Clinical |
| 31 | Kurian & Tripathi (2024) | Emotion recogn. | SSBD (75 videos) + YouTube | Children (web videos) / Ecol. | m_AutNet + WGAN | ACC 88.25% | Web |
| 32 | Manelis-Baram et al. (2025) | FER agreement | NADI ($N$=100) | ASD + TD 2–7 / Clinical | iMotions, FaceReader, Py-Feat | $r$=0.37–0.75 | Clinical |
| 48 | Kalantarian et al. (2019) | Emotion labelling | Game video ($N$=8 ASD) | ASD boys 8.5 y / Mobile game | Classifier ensemble | 56–94%[p] | Clinical |
| *Domain D2: Affect and therapeutic engagement (6 studies)* | | | | | | | |
| 4 | Lecciso et al. (2021) | Therapeutic interv. | Private ($N$=24) | ASD children / Therapy | OpenFace | $g$=0.52 | Clinical |
| 15 | Rudovic et al. (2018) | Affect / therapy | MIT private ($N$=35) | ASD children / Robot th. | PPA-net | ICC ∼60% | Clinical |
| 16 | Cicceri et al. (2026) | Pointing gestures | Private clinical | ASC children / Clinical | 4-level framework | mAP 88.6% | Clinical |
| 17 | Rudovic et al. (2019) | Active learning | MIT private ($N$=35) | ASD children / Robot th. | 4 LSTMs + Deep RL | ACC 82.3%, F1 52.6%[m] | Clinical |

*(Continues on next page)*

*(Continued from previous page)*

| # | Study | Clinical task | Dataset | Population / Setting | Architecture | Metrics | Data prov. |
|---|---|---|---|---|---|---|---|
| 34 | Shi et al. (2022) | Arousal / valence | Private in-home ($N$=4) | ASD mean 4.6 / In-home robot | XGBoost + s-DA | AUROC 0.92 / 0.86 | Clinical |
| 35 | Silva et al. (2021) | Real-time recogn. | Own non-ASD ($N$=43) | ASD $N$=6, 8–9 / Robot th. | SVM (RBF) | ACC 93.63%[q] | Non-clinical |
| *Domain D3: Emotion recognition by the child (6 studies)* | | | | | | | |
| 2 | He et al. (2019) | Eye-tracking | Private ($N$=42) | ASD preschoolers $n$=21 / Clinical | Eye-tracking | $r$=0.619 | Clinical |
| 39 | Voss et al. (2019) | Wearable interv. (RCT) | Trial data ($N$=71) | ASD 6–12 / Home | Glass FER classifier | VABS-II +4.58 | Clinical |
| 40 | Fan et al. (2025) | Emotion learning game | CAFE + FER-2013 | ASD $N$=12, 4–7 / School | CNN | Feedback ACC 97.6%[r] | Public gen. |
| 41 | Wall et al. (2025) | App interv., ET + EEG | Private ($N$=35) | ASD 15 + NT 20, 6–12 / ND | Eye-tracking + ERP | $Z$=−3.98 | Clinical |
| 45 | Carter Leno et al. (2020) | Attention to fear | Cohort ET ($N$=46) | ASD adolescents / ND | Eye-tracking | Assoc. with CU traits | Clinical |
| 51 | Bochet et al. (2021) | Emotional face expl. | Private ET ($N$=46) | ASD 24 + TD 22, 1.8–5.8 / ND | Eye-tracking | $p$=0.004 | Clinical |
| *Domain D4: Screening or diagnosis of ASD (12 studies)* | | | | | | | |
| 5 | Shaik et al. (2025) | ASD detection | Kaggle ASD | Children / Lab. | Ensemble CNN | ACC 84%, F1 84.2% | Web |
| 12 | Sariyanidi et al. (2025) | Autism pred. | CHOP clinical ($N$=42) | ASD + TD / Clinical | Dict. Learn. + SVM | Fig.[c] | Clinical |
| 13 | Hassan et al. (2025) | ASD level assess. | DREAM ($N$=61) | ASD children 3–6 / Robot th. | RF + XGBoost + ExtraT. | ACC 95.6%, F1 92.8%[n] | Public cur. |
| 19 | Alam et al. (2025) | ASD detection | Kaggle + YTUIA; UIFID test | Children / Lab. | ASD-UANet | ACC 96% | Web |
| 20 | Hwang et al. (2024) | DD screening | Private ($N$=89; 33 DD, 56 TD) | 34–77 months / Mobile | LSTM + 2D-FAN | ACC 88%, F1 80% | Clinical |
| 23 | Vedantham et al. (2025) | ASD detection | Kaggle | Children / Lab. | GWO-GEfficientNet | ACC 96.5%, F1 96.4% | Web |
| 24 | Hosney et al. (2024) | ASD detection | Kaggle (co-auth.)[d] | Children / Lab. | YOLOv8 + DCNN | ACC 97.2%, F1 93% | Web |
| 33 | Chen et al. (2024) | ASD identification | Private ($N$=81) | ASD 41 + TD 40, 4–6 / School | LSTM + SENet | ACC 88.74% | Clinical |
| 36 | Alvari et al. (2021) | Early smile markers | Home videos ($N$=33) | ASD 18 + TD 15, 6–12 m / Home | OpenFace AUs | MANOVA $p < 0.0001$ | Clinical |
| 37 | Carpenter et al. (2020) | Toddler screening | Private ($N$=104) | Toddlers (22 ASD) / Tablet | CV + LASSO logistic | AUC 0.62–0.73 | Clinical |
| 38 | Pliska et al. (2025) | Digital screening | Private ($N$=48) | ASD 24 + TD 24, 6–11 / Clinic, school | Decision trees | ACC 81.25% | Clinical |
| 49 | Han et al. (2022) | ASD identification | Private ($N$=90) | ASD 40 + TD 50, 3–6 / ND | MMSDAE fusion | ACC 95.56%, Sens. 92.5%[s] | Clinical |

## Section B. Contextual evidence outside the eligible set (9 studies)

Five infringe CE(3) (exclusively non-facial data), three infringe the age criterion (exclusively adult participants) and one infringes CI(3) (journal not indexed in SJR in the year of publication).

| # | Study | Clinical task | Dataset | Population | Setting | Architecture | Metrics | Data prov. |
|---|---|---|---|---|---|---|---|---|
| *Adult participants (3 studies)* | | | | | | | | |
| C1 | El Rhatassi et al. (2024) | Emotion recogn. | FER-2013 + Kaggle autism | Autistic adults | Lab. | CNN vs ViT | ACC 65–74% | Public gen. |
| C2 | El Rhatassi et al. (2025) | Real-time recogn. | CK+ + HeyGen AI | Autistic adults | Lab. | CNN×3 + LSTM + CC | ACC val. 53% | Mixed |
| C3 | Levy et al. (2024) | Wearable system | FER-2013 | ASD adults ($N$=10) | Ecol. | CNN + HW wearable | Usability | Public gen. |

*(Continues on next page)*

*(Continued from previous page)*

| # | Study | Clinical task | Dataset | Population | Setting | Architecture | Metrics | Data prov. |
|---|---|---|---|---|---|---|---|---|
| *Non-facial data only, CE(3) (5 studies)* | | | | | | | | |
| C4 | Nagae & Lee (2022) | Therapy + EDA | Private EDA ($N$=11) | DD children | Robot th. | EDA + wavelet + SVM | Conf. matrix | Clinical |
| C5 | Papaioannou et al. (2021) | EEG cogn. load | EEG ($N$=63) | ASD + ADHD adults | Clinical | MSE + PLSC | ANOVA | Clinical |
| C6 | Xu & Ju (2023) | VBM brain | ABIDE ($N$=203) | ASD 8–12 ($n$=98) | Clinical | VBM | GMV+ADOS corr. | Public cur. |
| C7 | O'Sullivan et al. (2023) | Speaker diariz. | Clin. trial audio ($N$=72) | ASD 5–45 | Home | Pyannote + xvector | F1 76.5% | Clinical |
| C8 | Alsubai (2025) | ASD detect. (tab.) | Kaggle screening, text, clinical | 12–36 m; adults 60–96 | Screening | Transformer MDNCT | ACC 94.7% | Web |
| *Journal not indexed in SJR, CI(3) (1 study)* | | | | | | | | |
| C9 | Akın-Bülbül et al. (2024) | ET screening | Private ET | Infants 18–36 m | Clinical | KNN + features | ACC 81.45% | Clinical |

# Section C. Longitudinal tracking audit and multi-label coding matrix

Section C.1 presents the systematic longitudinal tracking audit for all 36 eligible studies, evaluating the presence of intraindividual temporal series, progress outcomes, test-retest reliability, and responsiveness to change. Section C.2 presents the multi-label coding matrix, documenting the objective variable type, output class structure, and training representation for each study.

The principal finding confirmed by the audit is that no study in the eligible corpus reports session-to-session longitudinal tracking variables. Studies with multiple recorded sessions do not fill this gap: Rudovic et al. (2018, 2019) (35 children, repeated robot interactions) measure affect or engagement per session independently, and Hassan et al. (2025) (61 children, 3,121 sessions, DREAM dataset) use the session data to predict the ASD level and the ADOS score of each child; none quantifies intraindividual change trajectories, test-retest reliability, or responsiveness. Shi et al. (2022) (4 children, 19 in-home sessions in total) predict arousal and valence within each session with a chronological split by child, but do not quantify change across sessions. Lecciso et al. (2021), Voss et al. (2019), Fan et al. (2025) and Wall et al. (2025) use pre-post designs; they compare two time points at group level but do not report intraindividual session-by-session series. Longitudinal therapeutic monitoring therefore remains an open gap.

## C.1. Longitudinal tracking audit (36 eligible studies)

Columns: *Sessions*: number of distinct measurement points or sessions in the study design; *Subject unit*: unit of analysis (individual vs group); *Intraindiv. series*: intraindividual temporal series across sessions (Yes / No); *Progress outcomes*: session-to-session progress indicators (Yes / No); *Test-retest*: test-retest reliability coefficient reported (Yes / No); *Responsiveness*: sensitivity to change across time or intervention (Yes / No).

| # | Study | Sessions | Subject unit | Intraindiv. series | Progress outcomes | Test-retest | Responsiveness |
|---|---|---|---|---|---|---|---|
| *Domain D1 (12 studies)* | | | | | | | |
| 1 | Johnson et al. (2025) | 1 (cross-sect.) | Group | No | No | No | No |
| 8 | Radočaj & Mart. (2025) | 1 (cross-sect.) | Group | No | No | No | No |
| 9 | Washington et al. (2022) | 1 (cross-sect.) | Group | No | No | No | No |
| 10 | Sholikah et al. (2024) | 1 (usability test) | Individual[e] | No | No | No | No |
| 21 | Aljbori et al. (2025) | 1 (cross-sect.) | Group | No | No | No | No |
| 27 | Dapogny et al. (2019) | 1 (clinical game) | Group | No | No | No | No |
| 28 | Leo et al. (2018) | 1 (cross-sect.) | Group ($N$=17) | No | No | No | No |
| 29 | Kiejdo et al. (2025) | 99 recordings | Group | No | No | No | No |
| 30 | Wang et al. (2025) | 1 (cross-sect.) | Group | No | No | No | No |
| 31 | Kurian & Tripathi (2024) | 1 (web videos) | Group | No | No | No | No |
| 32 | Manelis-Baram et al. (2025) | 1 (ADOS session) | Group ($N$=100) | No | No | No | No |
| 48 | Kalantarian et al. (2019) | 1 (game session) | Group ($N$=8) | No | No | No | No |
| *Domain D2 (6 studies)* | | | | | | | |
| 4 | Lecciso et al. (2021) | 2 (pre-post) | Individual | Pre-post only[f] | No | ND | ND |
| 15 | Rudovic et al. (2018) | Multiple[h] | Individual | No[h] | ND | ND | ND |
| 16 | Cicceri et al. (2026) | 1 (clinical) | Group | No | No | No | No |
| 17 | Rudovic et al. (2019) | Multiple[h] | Individual | No[h] | ND | ND | ND |
| 34 | Shi et al. (2022) | 19 in-home sessions[t] | Individual ($N$=4) | No[t] | No | No | No |
| 35 | Silva et al. (2021) | 1 (pilot test) | Individual ($N$=6) | No | No | No | No |
| *Domain D3 (6 studies)* | | | | | | | |
| 2 | He et al. (2019) | 1 (cross-sect.) | Individual | No | No | No | No |
| 39 | Voss et al. (2019) | 2 (pre-post, RCT) | Group ($N$=71) | Pre-post only[f] | No | ND | ND |
| 40 | Fan et al. (2025) | 2 (pre-post, 3 weeks) | Group ($N$=12) | Pre-post only[f] | No | ND | ND |
| 41 | Wall et al. (2025) | 2 (pre-post, 2 weeks) | Group ($N$=35) | Pre-post only[f] | No | ND | ND |
| 45 | Carter Leno et al. (2020) | 1 (cross-sect.) | Group ($N$=46) | No | No | No | No |
| 51 | Bochet et al. (2021) | 1 (cross-sect.) | Group ($N$=46) | No | No | No | No |
| *Domain D4 (12 studies)* | | | | | | | |
| 5 | Shaik et al. (2025) | 1 (cross-sect.) | Group | No | No | No | No |
| 12 | Sariyanidi et al. (2025) | 1 (screening) | Group ($N$=42) | No | No | No | No |
| 13 | Hassan et al. (2025) | 3,121 sessions total | Group ($N$=61) | No[g] | ND | ND | ND |
| 19 | Alam et al. (2025) | 1 (cross-sect.) | Group | No | No | No | No |
| 20 | Hwang et al. (2024) | 1 (screening) | Group ($N$=89) | No | No | No | No |
| 23 | Vedantham et al. (2025) | 1 (cross-sect.) | Group | No | No | No | No |
| 24 | Hosney et al. (2024) | 1 (cross-sect.) | Group | No | No | No | No |
| 33 | Chen et al. (2024) | 1 (cross-sect.) | Group ($N$=81) | No | No | No | No |
| 36 | Alvari et al. (2021) | 1 (home videos) | Group ($N$=33) | No | No | No | No |
| 37 | Carpenter et al. (2020) | 1 (screening) | Group ($N$=104) | No | No | No | No |
| 38 | Pliska et al. (2025) | 1 (screening) | Group ($N$=48) | No | No | No | No |
| 49 | Han et al. (2022) | 1 (cross-sect.) | Group ($N$=90) | No | No | No | No |

## C.2. Multi-label coding matrix (36 eligible studies)

Columns: *Obj. variable*: type of target variable; *Output classes*: number of output categories; *Training repr.*: primary training representation modality; *Stratified val.*: stratified or subject-independent validation reported; *Multi-label*: whether the study reports multi-label (simultaneous assignment of ≥2 non-exclusive labels) as opposed to multi-class or binary single-label classification.

| # | Study | Obj. variable | Output classes | Training repr. | Stratified val. | Multi-label |
|---|---|---|---|---|---|---|
| *Domain D1 (12 studies)* | | | | | | |
| 1 | Johnson et al. (2025) | Multi-class emotion | 7 (basic emotions) | CNN / AffectNet+clinical | No | No |
| 8 | Radočaj & Mart. (2025) | Multi-class emotion | 4 (natural, fear, joy, anger) | Transformer / FERAC | Yes[l] | No |
| 9 | Washington et al. (2022) | Multi-class emotion | 7 (basic emotions) | CNN / CAFE subset | Yes | No |
| 10 | Sholikah et al. (2024) | Multi-class emotion | 7 (basic emotions) | CNN / FER-2013 | No | No |
| 21 | Aljbori et al. (2025) | Multi-class emotion | 6 (basic emotions) | 3D-CNN + LSTM / Kaggle | No | No |
| 27 | Dapogny et al. (2019) | Multi-class emotion | 4 (game expressions) | Rand. Forests / JEMImE | Yes | No |
| 28 | Leo et al. (2018) | Multi-class emotion | 7 + ASD prod. | HOG+CNN / CK+ + clinical | N/A | No |
| 29 | Kiejdo et al. (2025) | Multi-class emotion | 6 (basic emotions) | BiLSTM / FaceReader + biosignals | Yes (subject-disjoint) | No |
| 30 | Wang et al. (2025) | Multi-class emotion | 4 | CvT / face + speech | Yes (subject-wise) | No |
| 31 | Kurian & Tripathi (2024) | Multi-class emotion | 4 (happy, sad, neutral, meltdown) | CNN + WGAN / SSBD + YouTube | No[u] | No |
| 32 | Manelis-Baram et al. (2025) | Expression probabilities | N/A | Off-the-shelf FER / ADOS video | N/A | No |
| 48 | Kalantarian et al. (2019) | Multi-class emotion | 7 (6 emotions + neutral) | Commercial classifier ensemble / game frames | N/A | No |
| *Domain D2 (6 studies)* | | | | | | |
| 4 | Lecciso et al. (2021) | Continuous affect | N/A (AU intensity) | OpenFace / clinical | N/A | No |
| 15 | Rudovic et al. (2018) | Continuous affect | N/A (ICC scale) | PPA-net / MIT clinical | No[o] | No |
| 16 | Cicceri et al. (2026) | Multi-class gesture | 4 (pointing stages) | Spatial framework / clinical | No | No |
| 17 | Rudovic et al. (2019) | Multi-class engagement | 3 (low/med/high) | LSTM+RL / MIT clinical | Yes | No |
| 34 | Shi et al. (2022) | Binary arousal; binary valence | 2 + 2 | XGBoost / multimodal features | Yes (session-wise) | No |
| 35 | Silva et al. (2021) | Multi-class emotion | 6 | SVM / 3D landmarks + AUs | ND[q] | No |
| *Domain D3 (6 studies)* | | | | | | |
| 2 | He et al. (2019) | Correlation (continuous) | N/A | Eye-tracking | N/A | No |
| 39 | Voss et al. (2019) | Multi-class emotion | 8 | Embedded classifier / ND | N/A | No |
| 40 | Fan et al. (2025) | Multi-class emotion | 7 (FER-2013); 4 evaluated | CNN / CAFE + FER-2013 | No (image-level split) | No |
| 41 | Wall et al. (2025) | Group comparison (continuous) | N/A | Eye-tracking + ERP | N/A | No |
| 45 | Carter Leno et al. (2020) | Association (continuous) | N/A | Eye-tracking | N/A | No |
| 51 | Bochet et al. (2021) | Group comparison (continuous) | N/A | Eye-tracking | N/A | No |
| *Domain D4 (12 studies)* | | | | | | |
| 5 | Shaik et al. (2025) | Binary (ASD / TD) | 2 | CNN ensemble / Kaggle | Yes[o] | No |
| 12 | Sariyanidi et al. (2025) | Binary (ASD / TD) | 2 (graphical) | Dict. learn. / CHOP | Yes[o] | No |
| 13 | Hassan et al. (2025) | ASD level[g] | ND[g] | RF+XGBoost / DREAM | Yes[g] | No |
| 19 | Alam et al. (2025) | Binary (ASD / TD) | 2 | CNN ensemble / Kaggle+YTUIA | Yes | No |
| 20 | Hwang et al. (2024) | Binary (DD / TD) | 2 | LSTM+2D-FAN / clinical | Yes | No |
| 23 | Vedantham et al. (2025) | Binary (ASD / TD)[j] | 2 | CNN hybrid / Kaggle | No | No |
| 24 | Hosney et al. (2024) | Binary (ASD / TD) | 2 | YOLO+CNN / Kaggle | No | No |
| 33 | Chen et al. (2024) | Binary (ASD / TD) | 2 | LSTM + SENet / facial descriptors | Yes (by participant) | No |
| 36 | Alvari et al. (2021) | Group comparison (continuous) | N/A | OpenFace AUs | N/A | No |
| 37 | Carpenter et al. (2020) | Binary (ASD / non-ASD) | 2 | LASSO logistic / CV features | Yes (leave-one-out) | No |
| 38 | Pliska et al. (2025) | Binary (ASD / TD) | 2 | Decision trees / gaze + task | No[v] | No |
| 49 | Han et al. (2022) | Binary (ASD / TD) | 2 | MMSDAE / EEG + eye-tracking | Yes (10-fold, subject-indep.) | No |

**Notes.**

a. 3 ASD participants in Sholikah et al. (2024) involved in a usability test, not in metric computation.

b. TP 85% for Leo et al. (2018) corresponds to the CK+ public dataset of typically developing adults, not to the 17 ASD children.

c. Fig.: metrics reported in graphical format only, without numerically extractable values. Applies to Sariyanidi et al. (2025).

d. (co-auth.): Kaggle dataset created by a co-author of Hosney et al. (2024), constituting a documented conflict of interest.

e. $N$=3 ASD children; usability test only (see note a).

f. Lecciso et al. (2021) employs a pre-post design comparing two time points ($g$=0.52) but does not report intraindividual session-by-session series, test-retest reliability, or responsiveness. The same applies to the pre-post designs of Voss et al. (2019), a randomised controlled trial with 51% adherence, Fan et al. (2025) and Wall et al. (2025).

g. Hassan et al. (2025) use the 3,121 robot-enhanced therapy sessions of the 61 children in the DREAM dataset to predict the ASD level (classification) and the ADOS score (regression, RMSE 1.78, $R^2$ 0.74). Because the outcome is the severity of the condition and not engagement or affect during the session, the study is assigned to domain D4. The article states that "a stratified cross-validation technique was used to generate all of the training scores"; the number of folds and whether sessions are grouped by child are not available in the accessible text. It describes three ADOS-based levels (non-autism, autism spectrum disorder and autism) and notes that DREAM contains no participant without an ASD diagnosis, so the number of levels effectively classified and the cut-off points could not be confirmed and remain ND. The study does not quantify intraindividual change trajectories.

h. Rudovic et al. (2018, 2019) involve repeated interactions with a social robot ($N$=35); the design does not quantify intraindividual change trajectories, test-retest reliability, or responsiveness to intervention.

i. Aljbori et al. (2025) report 99.01% accuracy in their results table and 99.82% in the abstract; the table value is retained here.

j. Vedantham et al. (2025) classify autistic versus non-autistic facial images; an auxiliary facial expression dataset is relabelled into three classes (positive, negative, neutral) without a stated mapping to the autism labels.

k. The 67.8% reported by Johnson et al. (2025) corresponds to seven-class classification on AffectNet.

l. Radočaj and Martinović (2025) state an 80/20 stratified train-validation split, while the image counts they report are inconsistent (691/79 in one section and 615/155 in another).

m. Best values after adaptation in Rudovic et al. (2019): multimodal Q-learning with model-level fusion, three engagement levels (Tables 2 and 3 of the article), with training and test children kept separate.

n. The 95.6% accuracy of Hassan et al. (2025) corresponds to the boosted 3D biomarker approach for autism-level prediction; adding saliency maps, the full framework reports 97.36% accuracy and an F1-score of 95.56%.

o. Validation schemes verified in the full texts: Shaik et al. (2025) use stratified 5-fold cross-validation together with a stratified 80/20 split; Sariyanidi et al. (2025) use leave-one-out cross-validation by participant with a nested 5-fold inner loop for the SVM parameter; Rudovic et al. (2018) split each child's own data 40/20/40 into training, validation and test sets (personalised models), so validation is neither stratified nor subject-independent.

p. Kalantarian et al. (2019) label the frames of game videos of 8 boys with ASD (one session each) with commercial emotion classifiers combined in an ensemble, over 7 categories (6 emotions and neutral); the share of correctly labelled frames is 94% for disgust, 81% for neutral, 92% for surprise and 56% for scared, without significant improvement for angry and happy.

q. Silva et al. (2021) train on 43 participants without ASD (11 adults and 32 typically developing children) and report 93.63% with 10-fold cross-validation offline, without stating the partition unit, and 88.3% in real time with adults; the six children with ASD took part in a pilot test.

r. The 97.6% of Fan et al. (2025) is the accuracy of the real-time feedback of the system judged by specialists on 330 clips; the classifier was trained on data from children without ASD.

s. Han et al. (2022) report 86.67% accuracy for eye-tracking and 81.11% for EEG; the fused model reaches 95.56% accuracy, 92.5% sensitivity and 98% specificity under 10-fold subject-independent stratified cross-validation.

t. Shi et al. (2022) record 19 in-home sessions in total with a socially assistive robot and predict arousal and valence within each session with a chronological split by child; change across sessions is not quantified.

u. Kurian and Tripathi (2024) report a test accuracy of 88.25% on a hold-out set of 80 samples; separation of children between training and test is not reported.

v. Pliska et al. (2025) report no cross-validation or hold-out set; the accuracy is computed on the study sample, and the authors warn of overfitting.

1. The internal # column uses the corpus order shared with Supplementary Material S2 for eligible studies and the C prefix for contextual evidence. It does not correspond to bibliographic reference numbers, and gaps in the sequence are not numbering errors.
2. Retracted or withdrawn records do not appear, among them Güngör et al. (2025), Mahmood et al. (2025), Kumar et al. (2024) and Ahmad et al. (2024). Ahmad et al. (2024) was retracted on 17 August 2026 (*Healthc Technol Lett* 2026;13(1):e70095). All eight retracted or withdrawn records are listed in Supplementary Material S5 and documented in Section 2.7 of the manuscript.
3. Bal.ACC for Washington et al. (2022): balanced accuracy 79.1% on CAFE subset with human agreement $\geq$60%; full CAFE set value is 66.9%.
4. Data prov.: Non-clinical marks a training set built by the authors with participants without ASD.
5. Contextual studies in Section B were appraised with MMAT in Supplementary Material S2 (Section B) but do not participate in the aggregate MMAT distribution of the manuscript.